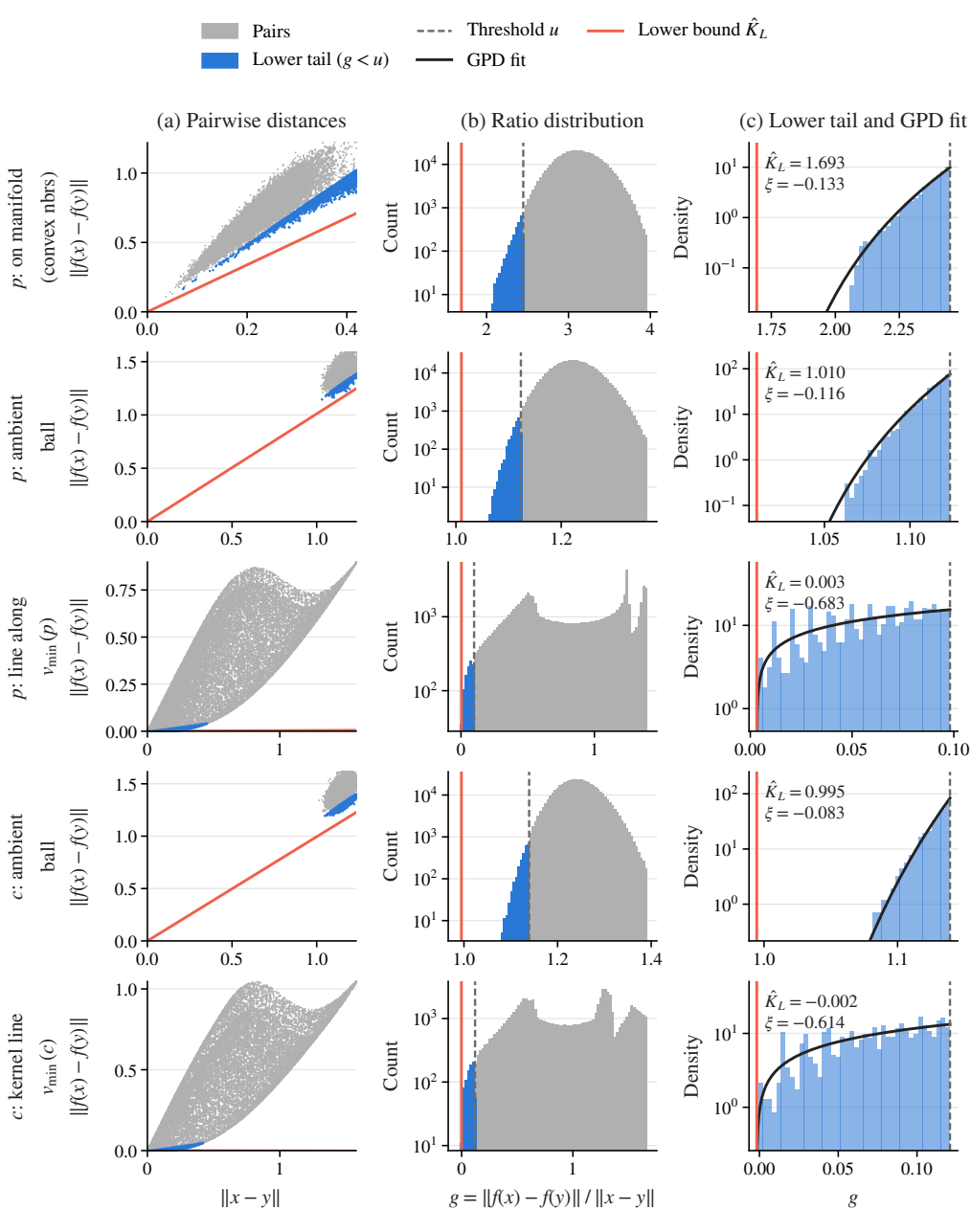

Pairs
Lower tail ($g < u$)
Threshold u
GPD fit
Lower bound $\hat{K}_L$
(a) Pairwise distances
(b) Ratio distribution
(c) Lower tail and GPD fit
p: on manifold (convex nbrs)
p: ambient ball
p: line along $v_{\min}(p)$
c: ambient ball
c: kernel line $v_{\min}(c)$
$\|f(x) - f(y)\|$
Count
Density
$\hat{K}_L = 1.693$
$\xi = -0.133$
$\hat{K}_L = 1.010$
$\xi = -0.116$
$\hat{K}_L = 0.003$
$\xi = -0.683$
$\hat{K}_L = 0.995$
$\xi = -0.083$
$\hat{K}_L = -0.002$
$\xi = -0.614$
$\|x - y\|$
$g = \|f(x) - f(y)\| / \|x - y\|$
g